MIL-L-48768 (PA)
19 June 1975

MILITARY SPECIFICATION

LITHIUM RESERVE BATTERY ELECTROLYTE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a type of electrolyte for use in lithium reserve type batteries (see 6.3).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification For
MIL-L-48765 - Lithium Hexafluoroarsenate
MIL-L-48766 - Lithium Tetrafluoroborate
MIL-M-48767 - Methyl Formate

STANDARDS

MILITARY

MIL-STD-129 - Marking for Shipping and Storage

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement function should be obtained from the procuring activity or as directed by the contracting officer).

FSC: 6810

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2.2 Other publications.-The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM-E70 - pH of Aqueous Solutions with the Glass Electrode
ASTM-E203 - Water Using Karl Fischer Reagent
ASTM-E260 - General Gas Chromatography Procedures
ASTM-E300 - Sampling Industrial Chemicals
ASTM-D445 - Kinematic Viscosity of Transparent and Opaque Liquids

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103).

3. REQUIREMENTS

3.1 Materials.-Materials used in the manufacture of the electrolyte shall conform with the specifications given below and shall be mixed in the proportions shown:

Materials	Applicable Specification	Proportions By Weight (see 6.3)
Lithium Hexafluoroarsenate	MIL-L-48765	18
Lithium Tetrafluoroborate	MIL-L-48766	1.72
Methyl Formate	MIL-L-48767	40.00

3.2 Properties.-The electrolyte shall conform to the following

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properties when tested in accordance with applicable paragraphs (see 6.5):

Property	Requirement	Applicable Paragraph
Color	Clear & Colorless	4.4.3.1
Insoluble Matter	$<$0.5%	4.4.3.2
Specific Gravity @ 25.0 ± 0.1°C	1.2850 to 1.3150	4.4.3.3
Viscosity @ 25.0 ± 0.1°C centistokes	1.3900 to 1.6100	4.4.3.4
Purity		
Water,ppm,max	300	4.4.3.5
Methanol,ppm,max	300	4.4.3.6
Formic Acid,ppm,max	100	4.4.3.6
Dimethyl Ether,ppm,max	100	4.4.3.6
Total Impurities,ppm, max	500	4.4.3.6
Refractive Index @ 25 ± 2°C	1.352 to 1.355	4.4.3.7
pH @ 25 ± 2°C	2.8 to 4.0	4.4.3.8
Diffussion Current, microamps	$<$20	4.4.3.9
Specific Conductance @ 25 ± 0.1°C	39 to 46	4.4.3.10

3.2 First article testing.-This specification makes provisions for first article testing. Requirements for the submission of first article samples by the contractor shall be as specified in the contract. (see 6.1)

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection and Standard Quality Assurance Provisions.-Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of Inspections.-The inspection requirements specified herein are classified as follows:

a. First Article Inspection (see 4.3)
b. Quality Conformance Inspection (see 4.4)

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4.3 First article inspection

4.3.1 Submission.-The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article shall consist of 1 gallon of electrolyte obtained by sampling as described in 4.4.2.1. The sample shall be obtained from a production lot which has been produced by the contractor using the same production processes, procedures and equipment as will be used in fulfilling the contract. All materials, shall be obtained from the same sources of supply as will be used in regular production.

4.3.2 Inspections to be performed.-The sample will be subjected by the Government to any or all of the examinations or tests specified in 4.4.3 of this specification.

4.3.3 Rejection.-See MIL-A-48078(PA)

4.4 Quality Conformance Inspection

4.4.1 Lot formation.-A lot shall consist of one batch of electrolyte produced by one manufacturer, in accordance with the same specification, or same specification revision, under one continuous set of operating conditions. Each lot shall consist of that quantity of electrolyte that has been subjected to the same unit chemical or physical mixing process intended to make the final product homogeneous. The product shall be submitted for inspection in accordance with MIL-STD-105.

4.4.2 Testing.-Testing and sampling shall be performed in a dry atmosphere.

4.4.2.1 Sampling.-Approximately 2 liters of electrolyte (see 6.4) shall be withdrawn from the lot to be sampled using ASTM Procedure E 300-70 for liquids and placing sample in a 2 liter glass flask. Prior to withdrawing sample, the lot shall be thoroughly mixed using a suitable agitator to suspend any insoluble solids present in the electrolyte. If the sample fails to meet any of the test requirements, the lot shall be rejected. The sample shall be subjected to the tests listed below:

Inspection/Requirement (see 3.2)	Defect Classification
Color	Major B
Insoluble Matter	Major B
Specific Gravity	Major B
Viscosity	Major B
Purity	Major B
Refractive Index	Major B
Diffussion Current	Major B
Specific Conductance	Major B

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4.4.2.2 Inspection Equipment.-The government reserves the right to inspect the contractor's equipment and determine that he has available and utilizes correctly, measuring and test equipment of the required accuracy and precision and that the instruments are of the proper type and range to make measurements of the required accuracy. Commercial inspection equipment, shall be employed where applicable for all tests and examinations specified in 4.4.3. The contractor is responsible for assuring proper calibration procedures are followed. Government approval of inspection equipment is required prior to its use for acceptance purpose.

4.4.3 Test methods and procedures.-(see 6.2) The tests in 4.4.3.3 thru 4.4.3.10 shall be performed using prescribed analytical procedures for replicate determinations given in standard analytical textbooks.

4.4.3.1 Color.-Visually observe the color of the liquid sample obtained in 4.4.2.1.

4.4.3.2 Insoluble matter.-Quantitatively filter the contents of the flask (4.4.2.1) through a clean, tared glass medium porosity filtering crucible. Use a small amount of methyl formate to aid in the transfer. Dry the crucible and contents in an oven maintained at 85 $\pm$ 5°C for 1 hour, cool in a desiccator and weigh. Calculate the Insoluble Matter as compared to the total $LiAsF_6$ and $LiBF_4$ according to the following equation:

$$\% \text{ Insoluble Matter} = \frac{\text{Wt Insoluble in Grams X 100}}{\text{Volume Sample in Liters x 429.22}}$$

4.4.3.3 Specific Gravity.-The specific gravity shall be determined by the following procedure:

4.4.3.3.1 Equipment:

a. Two calibrated pycnometers (see Figure 1) fitted with Thomas red rubber stoppers - cat. No. 1781-E93.

b. 25cc syringe fitted with a 12 inch needle.

c. Controlled temperature chamber: 25°C $\pm$ 0.1°C

d. 20 cc of solution to be tested.

4.4.3.3.2 Procedure.-Weigh the dried pycnometers (with stopper). Fill each pycnometer until the solution is at least half way up the neck employing the 25 cc syringe and 12 inch needle. The needle

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must be inserted well into the bulb of the pycnometer to insure that no bubbles are trapped in the neck. Place the pycnometers in the controlled temperature chamber and allow them to reach the temperature (25 ± 0.1°C). Record the volume in ml. Calculate the specific gravity employing the following equation:

$$SG = \frac{\text{Weight of solution}}{\text{Volume of solution}} \times \frac{1}{0.99707 \text{ g/cc}}$$

Add the values obtained and divide by two to obtain the specific gravity of the solution being tested.

4.4.3.4 Viscosity.-Determine the viscosity of the electrolyte in accordance with ASTM D-445, Ostwald Type at 25.5 ± 0.1°C.

4.4.3.5 Water.-Determine the water content of the sample in accordance with Method ASTM-E203.

4.4.3.6 Methanol, Formic Acid, Dimethyl Ether.-Determine the impurities in the electrolyte in accordance with ASTM-E260 with following additions:

a. Injector port - glass wood packed
b. Column - Chromosorb 101, 4 ft. x 1/8" diameter.
c. Detector - Flame ionization (methanol and dimethyl ether)
d. Thermal Conductivity (Formic Acid)
e. Carrier Gas - Helium at 20 cc/min.

Calculate the total impurities by adding values found in 4.4.3.6 to the value for water in 4.4.3.5.

4.4.3.7 Refractive Index.-Determine the refractive index of the electrolyte using a Fisher Refractometer or equivalent in accordance with the manufacturers procedural instructions of 25 ± 2°C.

4.4.3.8 pH.-Determine the pH of the electrolyte in accordance with method ASTM-E70 at 25 ± 2°C except that electrolyte shall be diluted in the ratio of one part electrolyte to nine parts of distilled water and pH measured immediately.

4.4.3.9 Diffusion Current.-Determine the diffusion current of the electrolyte using standard polarographic techniques in

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accordance with procedures in a standard or reference textbook with following modifications:

a. Reference electrode - Lithium
b. Auxiliary electrode - Lithium
c. Working electrode - Dropping Mercury
d. Measured at 0.0 to -1.75 volts versus the hydrogen electrode.

4.4.3.10 Specific Conductance - Determine the specific conductance of the electrolyte at 25.0 ± 0.1°C as follows:

4.4.3.10.1 Equipment.-

a. Beckman Conductivity Bridge or equivalent (1)
b. Decade Capacitors (1)
c. Controlled temperature chamber: 25°C ± 1°C (1)
d. Calibrated conductivity cells of 5 ml volume air tight seals (2)
e. Solution to be tested (10 cc)
f. 10 cc syringe (1)

Procedure: Transfer 3 ml of the solution to be tested to each conductivity cell. (The conductivity cells should be dried thoroughly prior to use.) Place the cells in the controlled temperature chamber and allow them to reach temperature (25 ± 0.1°C). Record three separate readings in ohms employing a frequency of 10,000 Hertz and average the values. Calculate the specific conductance (L_S) according to the following equations:

$L_S = K/R$ Where: K = cell constant
R = resistance in ohms

Add the two values obtained and divide by two to obtain the specific conductance of the solution being tested.

5. PREPARATION FOR DELIVERY

5.1 Packing.-(see 6.5)

5.1.1 Level A and Level C.-Unless otherwise specified, the electrolyte shall be packed in accordance with manufacturer's commercial practice to assure acceptance by common carrier for safe delivery at first destination for immediatex use.

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5.2 Marking.-Unless otherwise specified, the shipping containers shall be marked in accordance with MIL-STD-129, and shall include the lot number.

6. NOTES

6.1 Ordering data.-See MIL-A-48078(PA).

6.2 Prior approval of the Contracting Officer is required for use of equivalent test methods. A description of the proposed method should be submitted thru the Contracting Officer to: Commander, ATTN: SARPA-QA-A-P, Picatinny Arsenal, Dover, New Jersey 07801. This description should include but not be limited to the procedures used, the accuracy and precision of the method, test data to demonstrate the accuracy and precision and drawings of any special equipment required.

6.3 The electrolyte is 2 Molar lithium hexafluoroarsenate and 0.4M lithium tetrafluoroborate in methyl formate. The electrolyte contains the following properties by weight:

Li AsF6.....18 Parts
LiBF4.....1.72 parts
MF.....40 parts

6.4 Before test samples are drawn, the material shall be thoroughly mixed to assure uniformity. Every precaution should be taken to prevent loss of volatile constituents or introduction of moisture prior to and during drawing of test sample.

6.5 Handling and Storage.-The electrolyte should be stored at room temperature with strips of pure lithium metal in sealed clear glass bottles, which are in addition stored in sealed polyethylene bags containing an appropriate desiccant.

Custodian:
Army-PA

Preparing Activity:
Army-PA

Project Number: 6810-AA68

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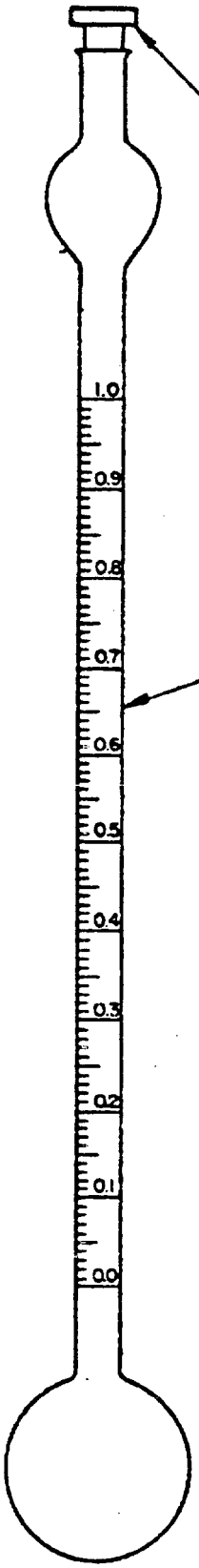


FIGURE 1
PYCNOMETER FOR SPECIFIC GRAVITY DETERMINATION

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